



WTVB02-485 Data Sheet & User Manual



Product Specifications:

Model: WTVB02-485

Description: Vibration sensor

Contents

1. Product Overview	1
2. Parameter indicators	1
2.1. Basic parameters	1
2.2. Parameter Description	2
2.3. Product size	2
2.4. Parameter comparison	3
2.5. Electrical parameters	3
3. Hardware connection method	4
3.1. Wiring method	4
3.2. Installation	4
4. How to use the software	5
4.1. PC software download	5
4.2. Connect to the PC software	5
4.2.1. Automatic Search	6
4.2.2. Manual Connection	7
4.3. Graph	8
4.4. Scatter plots	8
5. Software Configuration	9
5.1. Restore settings	9
5.2. Reboot	9
5.3. Communication rate	10
5.4. Device Address	10
5.5. Record Data	11
6. Communication Protocol	11

6.1. Read register format	11
6.2. Write register format	12
6.3. Register Address Table	13
6.4. Register Description	21
6.4.1. SAVE (Save/Reboot/Reset to Factory)	21
6.4.2. BAUD (serial port baud rate)	21
6.4.3. IICADDR (device address)	22
6.4.4. YYMM~MS (on-chip time)	23
6.4.5. AX~AZ (acceleration)	24
6.4.6. VX~VZ (vibration speed)	25
6.4.7. Reserve	25
6.4.8. TEMP (temperature)	26
6.4.9. DX~DZ (vibration displacement)	26
6.4.10. HZX ~ HZZ (vibration frequency)	27
6.4.11. Displacement range mode	28
6.4.12. Sampling rate	29
6.4.13. Original Acceleration FIFO	29
6.4.14. X-axis vibration simulation quantity	30
6.4.15. Y-axis vibration simulation quantity	31
6.4.16. XZ-axis vibration simulation quantity	32
6.4.17. XYZ axis fault diagnosis	33
6.4.18. X-axis focus and acc spectrum energy	34
6.4.19. Y-axis focus and acc spectrum energy	35
6.4.20. Z-axis focus and acc spectrum energy	37

1. Product Overview

1. The module has its own voltage stabilization circuit, the operating voltage is 9V~36V, and the connection is easy.
2. The use of advanced digital filtering technology can effectively reduce measurement noise and improve measurement accuracy.
3. At the same time, provide users with various PC softwares, instructions for use, and development manuals.
4. Support 485 interface. It is convenient for users to choose the best connection method. The serial port rate is adjustable from 4800bps to 230400bps.
5. Application areas: It can be widely used in bearing vibration measurement and real-time monitoring of rotating machinery such as submersible pumps, fans, steam turbines, coal mills, oxygen generators, generators, centrifuges, compressors, water pumps, motors, etc.
6. Three-axis displacement, three-axis velocity, and three-axis frequency outputs enable users to comprehensively measure vibration and shock, determining whether the measured object (motor or water pump) is damaged. If a machine malfunction is caused by factors such as bearing wear, cracking, poor dynamic balance, and misalignment, the vibration sensor can detect the fault in advance and issue an early warning, preventing the machine from continuing to operate under adverse conditions and causing damage, which could lead to economic losses.
7. Multiple installation methods: magnetic connection, threaded connection. Firm and stable, easy to install and disassemble.
8. Stud bolts with positive and negative threads: stainless steel hexagonal stud bolts with positive and negative threads, double-ended screws, studs, left-hand and right-hand bidirectional screws

2. Parameter indicators

2.1. Basic parameters

Parameter	Condition	Minimum	Default	Maximum
Communication interface	485 interface	4800bps	9600bps	230400bps
Output content	On-chip time, 3-axis acceleration, 3-axis vibration velocity, 3-axis vibration displacement, 3-axis vibration frequency, temperature			

Range	Vibration speed: 0~600mm/s, vibration displacement: 0~60000um, vibration frequency 5-2000hz			
Accuracy	<FS±4%			
Operating temperature		-40℃		85℃
Storage temperature		-40℃		85℃
impact resistant	20000g			
Protection level	IP67			

2.2. Parameter Description

1. Output content

- Vibration velocity: the speed at which the vibration point of an object moves when it vibrates
- Vibration displacement: Amplitude, the jitter amplitude of the vibration point when the object vibrates
- Vibration frequency: the number of times an object vibrates per unit time when it vibrates
- Temperature: Chip operating temperature, not external temperature

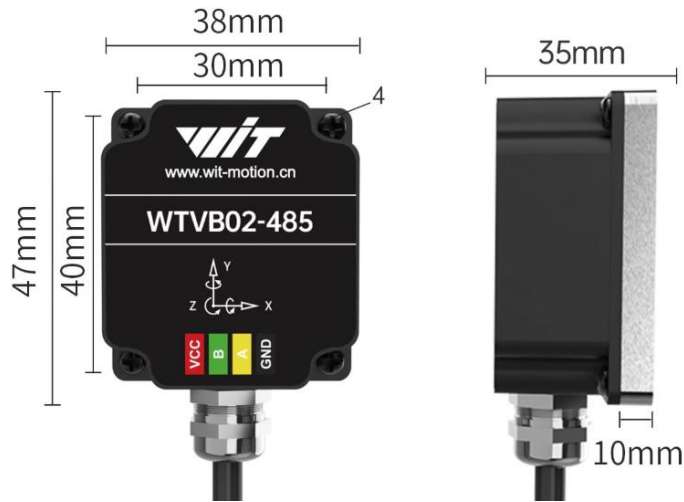
2. Accuracy

<FS±4%: indicates 4% of the measuring range. Taking vibration displacement as an example, when the actual displacement value is 300um, the data error is 4% of 300um. When the actual value is 3000um, the data error is 4% of 3000um.

Note: The product has requirements for vibration frequency and is not suitable for low-frequency vibration scenarios. The vibration frequency must be at least 5Hz to ensure data reliability.

2.3. Product size

WTVB02-485 product dimensions:



2.4. Parameter comparison

Product Appearance					
Model		WTVB02-485	WTVB01-485	WTVB05	WTVB01-BT50
Output	Vibration acceleration	✓	✓	✓	✓
	Vibration speed	✓	✓	✓	✓
	Vibration displacement	✓	✓	✓	✓
	Vibration frequency	✓	✓	✓	✓
	Chip Time	✓	✓	✓	✓
	Chip temperature	✓	✓	Ambient temperature	✓
Output rate	Normal mode (1-200Hz)	Send a command and return once(1-200Hz)			1-100Hz (default 20Hz)
Range	Vibration acceleration	±16G	±16G	±16G	✓
	Vibration displacement	0~60000um	0~30000um	0~60000um	0~30000um
	Vibration speed	0~600mm/s	0~600mm/s	0~600mm/s	0~600mm/s
	Vibration frequency	5-2000Hz	5-1000Hz	5-2000Hz	5-100Hz
Accuracy		<F.S±4%	<F.S±4%	<F.S±4%	<F.S±5%
Chip sampling rate		16KHz	1KHz	16KHz	1KHz
Communication interface		RS485	RS485	RS485/CAN	BLE5.0/TYPER-C
Communication Protocol		Modbus protocol	Modbus protocol	Modbus protocol CAN protocol	BLE5.0 protocol
Baud rate	RS485	4800bps-230400bps (default 9600bps)			115200bps (fixed)
	CAN	✓	✓	3K-1M(default 250K)	✓
Temperature	Operating temperature	-40°C~85°C			-20°C~+60°C
	Storage temperature	-40°C~85°C			
Waterproof grade		IP67			Not waterproof
Operating voltage		9-36V			5V/3.7V(Battery voltage)
Working current		8mA		12mA (12V)	5mA(3.7V)
Weight		182g		67.5g	20g
Size		47*38*35mm			51.5*36.1*15mm

2.5. Electrical parameters

Parameter	Condition	Minimum	Default	Maximum
-----------	-----------	---------	---------	---------

Supply voltage		9V	12V	36V
Working current	Working (12V)		8mA	

3. Hardware connection method

3.1. Wiring method

Wiring	RED	YELLOW	GREEN	BLACK
	VCC9-36V Positive pole of the power supply	A	B	GND Negative pole of the power supply



3.2. Installation

Installation Options

Magnetic attachment or threaded mount
Super stable, easy to install and remove



Magnetic Attachment

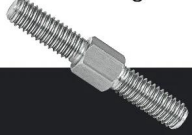
Magnet attaches to the device to detect vibration
Super suction up to 70kg.



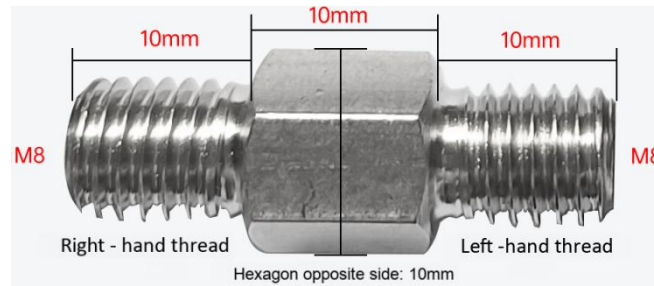


Threaded Mount

Default thread size is M8*1.25mm.
Stable threaded mounting



Thread size



4. How to use the software

4.1. PC software download

1. CH340 driver download link:

https://drive.google.com/file/d/1JidopB42R9EsCzMAYC3Ya9eJ8JbHapRF/view?usp=drive_link

2. Witmotion software download link:

https://drive.google.com/file/d/10xysnkuyUwi3AK_t3965SLr5Yt6YKE-u/view?usp=drive_link

4.2. Connect to the PC software

to the computer via a USB adapter , open Witmotion.exe , and after installing the CH340 driver, you can query the corresponding port number in the device manager, as shown in the figure:



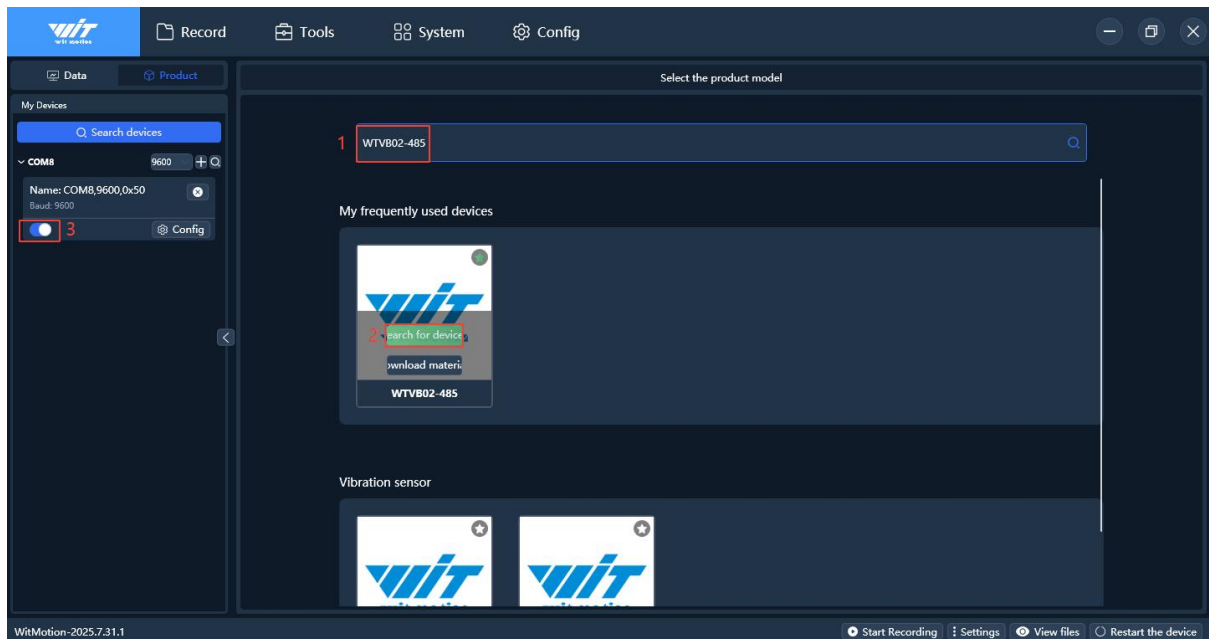
After the device is connected, open the Witmotion.exe software.



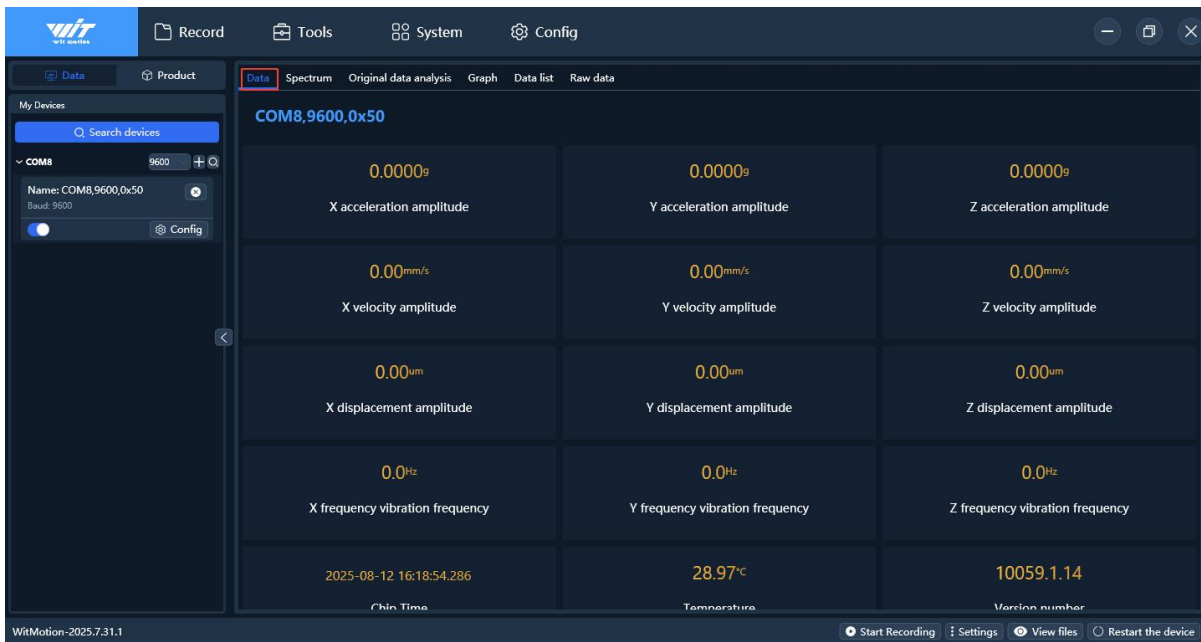
名称	修改日期	类型	大小
AutoUpdateApp	2024/9/6 星期五 17:21	文件夹	
Bin	2025/4/8 星期二 12:05	文件夹	
Config	2025/4/29 星期二 14:...	文件夹	
Plugins	2024/9/6 星期五 17:21	文件夹	
Record	2025/4/29 星期二 14:...	文件夹	
Temp	2024/9/6 星期五 17:24	文件夹	
1.txt	2025/4/18 星期五 19:...	文本文档	1 KB
debug.log	2025/4/18 星期五 19:...	文本文档	1 KB
WitMotion.exe	2025/4/16 星期三 17:...	应用程序	1,543 KB
WitMotion.exe.config	2025/4/16 星期三 17:...	CONFIG 文件	19 KB
WitMotion.pdb	2025/4/16 星期三 17:...	PDB 文件	424 KB

4.2.1. Automatic Search

1. Input sensor model WTVB02-485.
2. Click Search for device.
3. Check the serial port number.



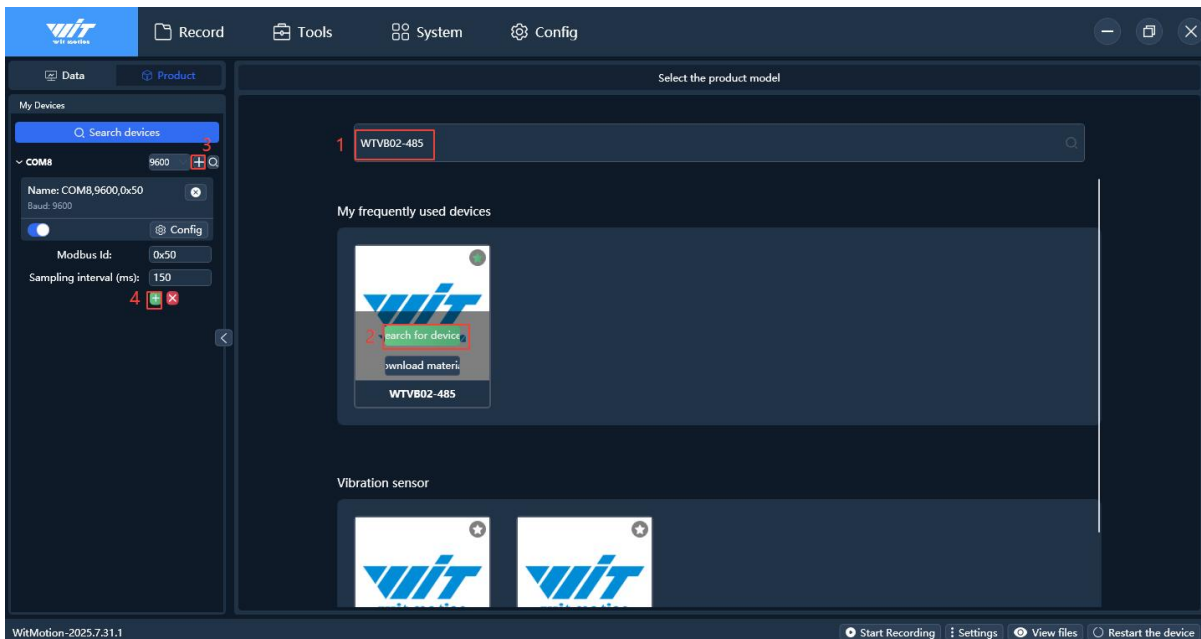
After the connection is successful, the data will be displayed on the software, as shown below:



4.2.2. Manual Connection

1. Input sensor model WTVB02-485.
2. Click Search for device.
3. Enter the corresponding ID (the default is 0x50) and click the + sign to add the device.

As shown in the figure below:



4.3. Graph

Click on the graph to see graphs of vibration speed, vibration angle, vibration displacement and temperature (normal mode);



4.4. Scatter plots

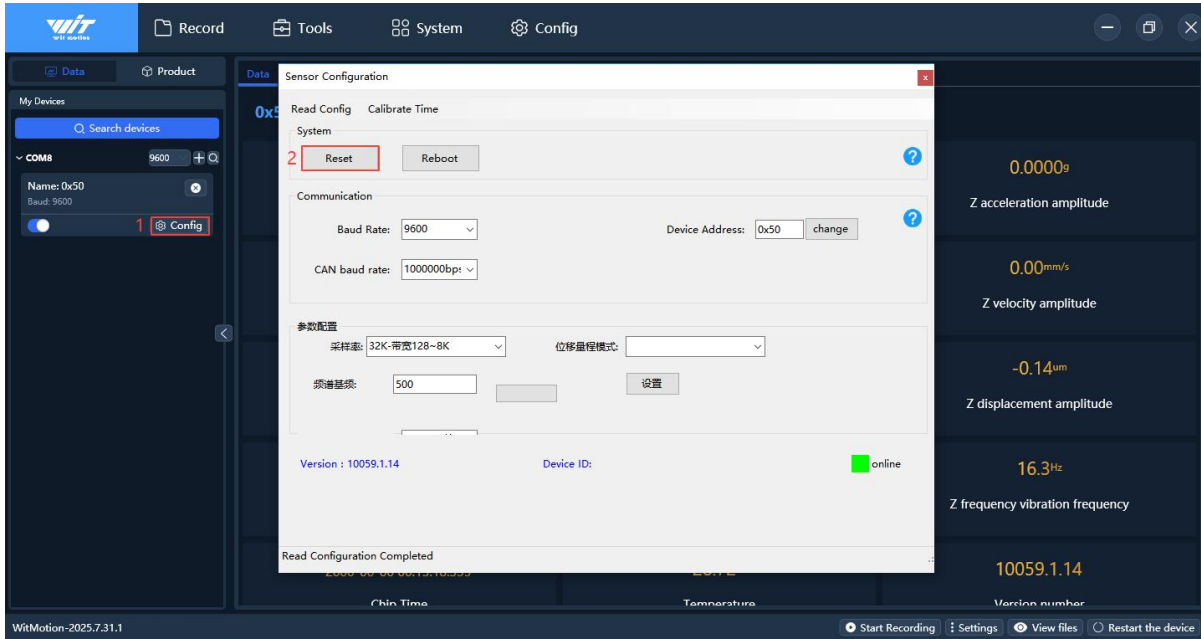
Vibration displacement scatter plot.



5. Software Configuration

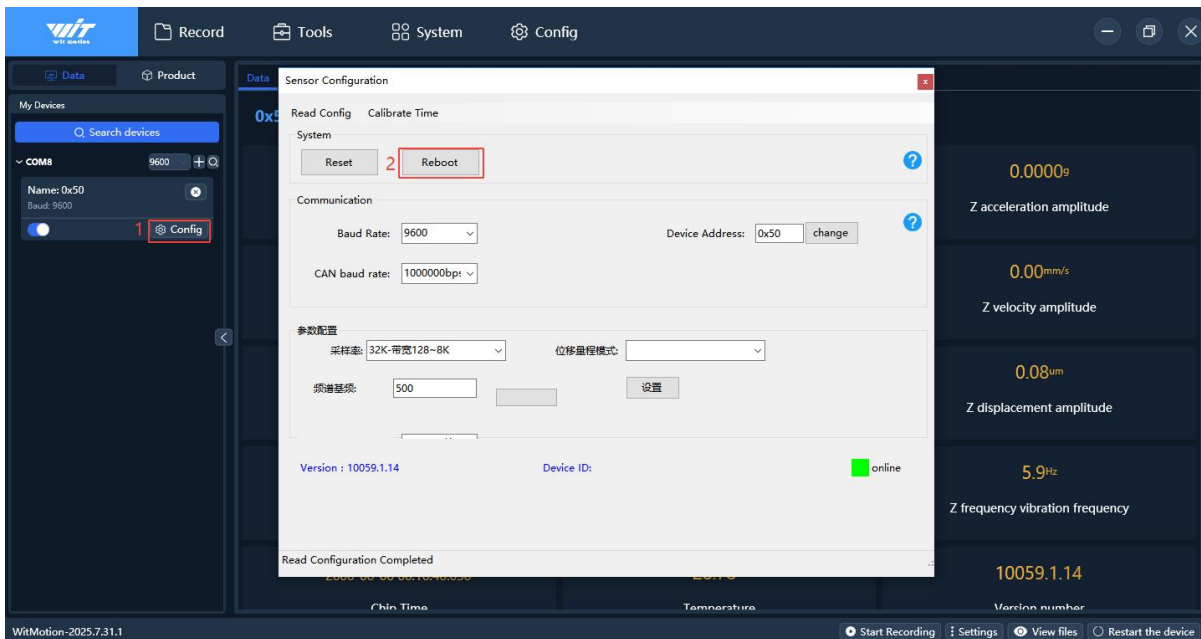
5.1. Restore settings

Click "Config" and then click "Reset" in the sensor configuration interface to restore the factory settings, as shown below:



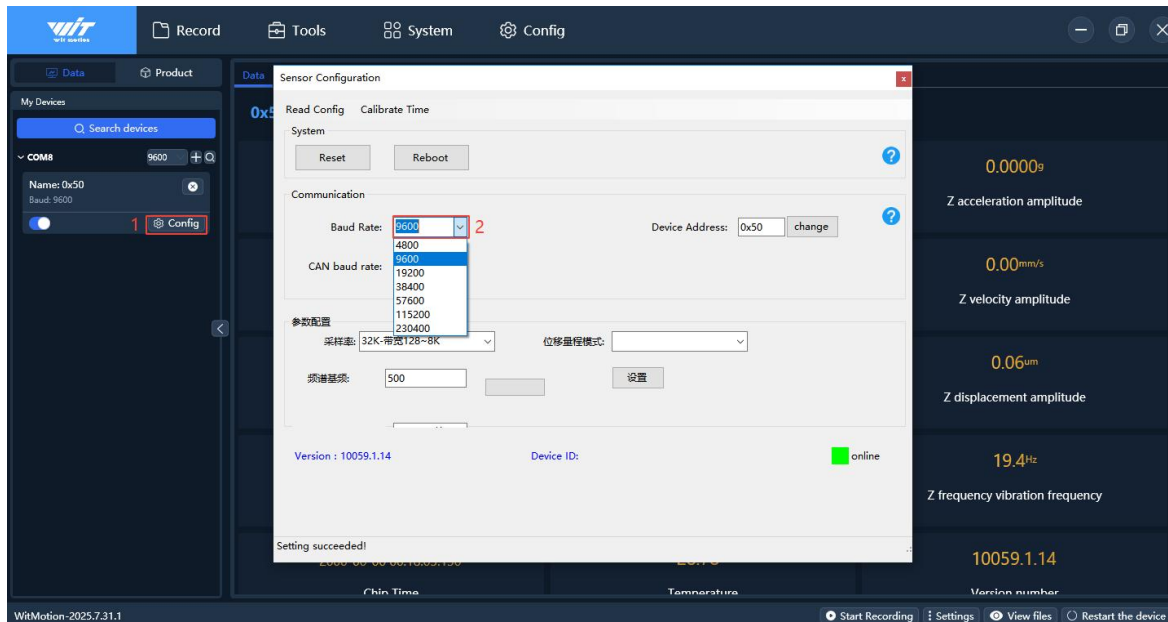
5.2. Reboot

Click "Config" and then click "Reboot" in the sensor configuration interface to restart the sensor, as shown below:



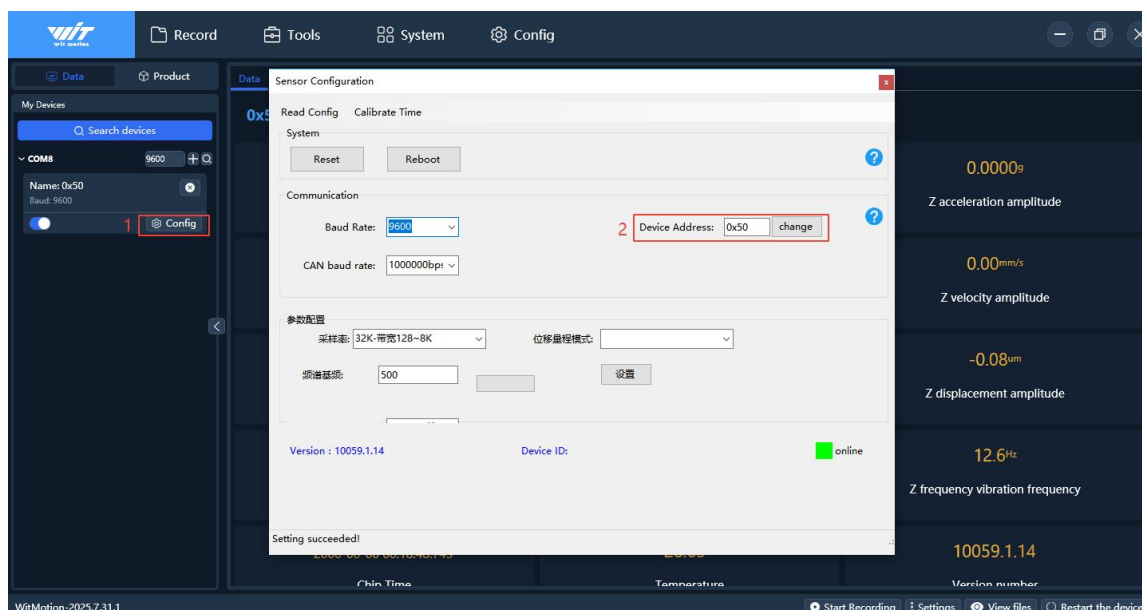
5.3. Communication rate

Click "Config" and click the "Baud Rate" drop-down menu in the sensor configuration interface. Select the serial port baud rate to be modified to change the current serial port baud rate (the default serial port baud rate is 9600). The serial port baud rate can be 4800, 9600, 19200, 38400, 57600, 115200, and 230400, as shown in the figure below:



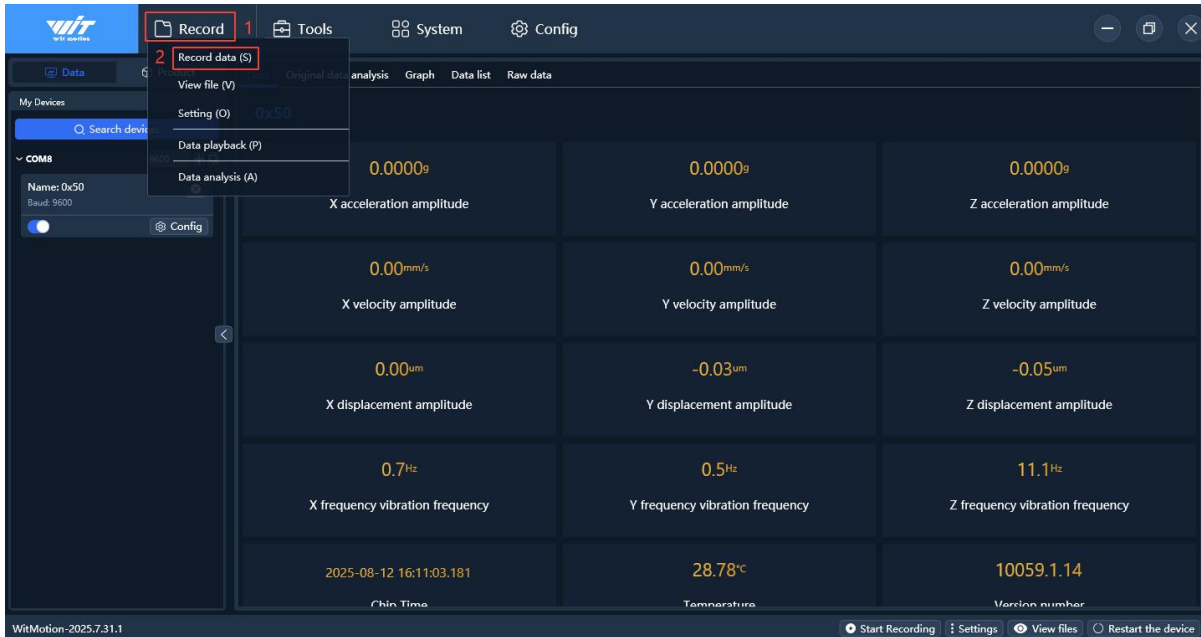
5.4. Device Address

Click "Config" and click the "Device Address" input box in the sensor configuration interface. Enter the Device Address and click Set to change the Device Address (the default Device Address is 0x50). The Device Address ranges from 0x00 to 0x7F, as shown in the figure:



5.5. Record Data

Click "Record" and click "Record data" to record the output data of the sensor.



6. Communication Protocol

Protocol: MODBUS protocol

Level: 485 level (default baud rate: 9600bps)

6.1. Read register format

- The data is sent in hexadecimal format, not ASCII.
- Each register address, register count, and data read are represented by two bytes. The high and low bits of the register address are represented by ADDR_H and ADDR_L, the high and low bits of the register count are represented by LEN_H and LEN_L, and the high and low bits of the data read are represented by DATA_{1H} and DATA_{1L}.
- The last two digits of the read instruction are standard CRC check digits.

Command sending

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	Read the high 8 bits of the length	Read the lower 8 bits of the length	Check digit high 8 bits	Check digit lower 8 bits
ID	0x03 (Read)	ADDR _H [15:8]	ADDR _L [7:0]	LEN _H [3:08 PM]	LEN _L [7:0]	CRCH[15:8]	CRCL[7:0]

Data return

Modbus Address	Function code	Read length	High 8 bits of data	Data lower 8 bits	High 8 bits of data	Data lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x03 (Read)	LEN[7:0]	DATA1H[15:8]	DATA1L[7:0]	...	...	DATANH	DATANL	CRCH[15:8]	CRCL[7:0]

6.2. Write register format

- The data is sent in hexadecimal format, not ASCII.
- Each register address and write data are represented by two bytes. The high and low bits of the register address are represented by ADDR_H and ADDR_L, and the high and low bits of the write data are represented by DATA_H and DATA_L.

Command sending

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x06 (Write)	ADDRH[15:8]	ADDRL[7:0]	DATAH[15:8]	DATAL[7:0]	CRCH[15:8]	CRCL[7:0]

Data return

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x06 (Write)	ADDRH[15:8]	ADDRL[7:0]	DATAH[15:8]	DATAL[7:0]	CRCH[15:8]	CRCL[7:0]

Note:

The instruction writing operation needs to be performed in three steps.

The first step is to unlock 0x50 0x06 0x00 0x69 0xB5 0x88 0x22 0xA1. The unlocking will take effect within ten seconds.

The second step is to send the instructions that need to be modified.

The third step is to save the instruction 0x50 0x06 0x00 0x00 0x00 0x84 0x4B.

6.3. Register Address Table

Reg Addr	Symbol	Meaning	Units and Explanations
0x00	SAVE	Save/Restart/Restore to Factory	
0x04	BAUD	Serial port baud rate	
0x1A	IICADDR	Device Address	
0x30	YYMM	Month Year	
0x31	DDHH	Time	
0x32	MMSS	Seconds and minutes	
0x33	MS	millisecond	
0x34	AX	X-axis acceleration	g
0x35	AY	Y-axis acceleration	g
0x36	AZ	Z- axis acceleration	g
0x37	AXF	X-axis acceleration amplitude	g
0x38	AYF	Y-axis acceleration amplitude	g
0x39	AZF	Z-axis acceleration amplitude	g
0x3A	VX	X-axis vibration speed	mm/s
0x3B	VY	Y-axis vibration speed	mm/s
0x3C	VZ	Z-axis vibration speed	mm/s
0x40	TEMP	Chip temperature	°C
0x41	DX	X-axis vibration displacement	mm
0x42	DY	Y-axis vibration displacement	mm

0x43	DZ	Z-axis vibration displacement	mm
0x44	HZX	X-axis vibration frequency	Hz
0x45	HZY	Y-axis vibration frequency	Hz
0x46	HZZ	Z-axis vibration frequency	Hz
0x47	CFX	X-axis acceleration peak factor	Peak value to RMS ratio, measuring the degree of signal abrupt change g
0x48	KX	X-axis acceleration kurtosis value	Detecting impact components or abnormal vibrations g
0x49	AAVGX	The absolute average of X-axis acceleration	The average of the absolute values of vibration, measuring overall intensity g
0x4A	VARX	X-axis acceleration variance	Reflects the magnitude of signal fluctuations g
0x4B	RRAX	X-axis acceleration root square magnitude (RMS)	The standard index of energy intensity g
0x4C	WIX	X-axis acceleration waveform index	The ratio of RMS to the absolute mean reflects the waveform shape g
0x4D	PIX	Peak X-axis acceleration	The ratio of peak value to absolute average value g
0x4E	PCX	X-axis acceleration pulse coefficient	
0x4F	SKX	X-axis acceleration skewness	The measure of signal symmetry (positive or negative bias) g

0x50	VRMSX	X-axis velocity RMS value	Velocity signal energy intensity (mm/s)
0x51	VKX	X-axis velocity kurtosis value	Speed signal impact detection g
0x52	DRMSX	X-axis displacement RMS value	Measurement of displacement change intensity (mm)
0x53	CFY	Y-axis acceleration peak factor	The peak value to RMS ratio measures the degree of signal abrupt change g
0x54	KY	Y-axis acceleration kurtosis value	Detection of impact components or abnormal vibrations g
0x55	AAVGY	Absolute average Y-axis acceleration	The average of the absolute values of vibrations measures the overall intensity g
0x56	VARY	Y-axis acceleration variance	Reflects the magnitude of the signal fluctuation range g
0x57	RRAY	Y-axis acceleration root square magnitude (RMS)	The standard index of energy intensity g
0x58	WIY	Y-axis acceleration waveform index	The ratio of RMS to the absolute mean reflects the waveform shape
0x59	PIY	Y-axis acceleration peak index	The ratio of peak value to absolute average value g
0x5A	PCY	Y-axis acceleration pulse coefficient	The ratio of peak value to average amplitude
0x5B	SKY	Y-axis acceleration skewness	Measuring signal symmetry (positive or negative bias) g

0x5C	VRMSY	Y-axis velocity RMS value	Velocity signal energy intensity mm/s
0x5D	VKY	Y-axis velocity kurtosis value	Speed signal impact detection g
0x5E	DRMSY	Y-axis displacement RMS value	Measurement of displacement change intensity mm
0x5F	CFZ	Z-axis acceleration peak factor	The peak value to RMS ratio measures the degree of signal abrupt change g
0x60	KZ	Z-axis acceleration kurtosis value	Detection of impact components or abnormal vibrations g
0x61	AAVGZ	Absolute average Z-axis acceleration	The average of the absolute values of vibrations measures the overall intensity g
0x62	VARZ	Z-axis acceleration variance	g reflects the magnitude of the signal fluctuation range
0x63	RRAZ	Z-axis acceleration root square magnitude (RMS)	The standard index of energy intensity g
0x64	WIZ	Z-axis acceleration waveform index	The ratio of RMS to the absolute mean reflects the waveform shape
0x65	PIZ	Z-axis acceleration peak index	The ratio of peak value to absolute average value g
0x66	PCZ	Z-axis acceleration pulse coefficient	The ratio of peak value to average amplitude
0x67	SKZ	Z-axis acceleration skewness	The measure of signal symmetry (positive or negative bias) g

0x68	VRMSZ	Z-axis speed RMS value	Velocity signal energy intensity mm/s
0x69	VKZ	Z-axis velocity kurtosis value	Speed signal impact detection g
0x6A	DRMSZ	Z-axis displacement RMS value	Measurement of displacement change intensity mm
0x6B	ERRX	X-axis fault diagnosis	
0x6C	ERRY	Y-axis fault diagnosis	
0x6D	ERRZ	Z-axis fault diagnosis	
0x6E	AX_ENE RGY_POI NT1	X-axis focus 1: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals at quarter harmonics
0x6F	AX_ENE RGY_POI NT2	X-axis focus point 2: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals using half harmonics
0x70	AX_ENE RGY_POI NT3	X-axis focus point 3: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (first harmonic)
0x71	AX_ENE RGY_POI NT4	X-axis focus point 4: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (second harmonics)
0x72	AX_ENE RGY_POI NT5	X-axis focus point 5: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (third harmonic)
0x73	AX_ENE RGY_POI NT6	X-axis focus point 6: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fourth harmonic)
0x74	AX_ENE RGY_POI NT7	X-axis focus point 7: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fifth harmonic)
0x75	AX_ENE RGY_POI NT8	X-axis focus point 8: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (sixth harmonic)
0x76	AX_ENE	X-axis focus band 1 acceleration	Energy monitoring of acceleration signals in

	RGY_AR EA1	spectrum energy	the frequency band from 0 to 1/2 octave.
0x77	AX_ENE RGY_AR EA2	X-axis focus band 2 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 1/2 to 1 harmonic.
0x78	AX_ENE RGY_AR EA3	X-axis focus on the acceleration spectrum energy of band 3	Energy monitoring of acceleration signals in the frequency band (1st to 2nd harmonic)
0x79	AX_ENE RGY_AR EA4	X-axis focus on band 4 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band at 2-3 harmonics
0x7A	AX_ENE RGY_AR EA5	X-axis focus on the acceleration spectrum energy of band 5	Energy monitoring of acceleration signals in the frequency band at 3-4 harmonics
0x7B	AX_ENE RGY_AR EA6	X-axis focus on the acceleration spectrum energy of band 6	Energy monitoring of acceleration signals in the frequency band at 4-5 harmonics
0x7C	AX_ENE RGY	Total energy of X-acceleration spectrum	The sum of acceleration energy across all frequency ranges
0x7D	AY_ENE RGY_POI NT1	Y-axis focus point 1: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals at quarter harmonics
0x7E	AY_ENE RGY_POI NT2	Y-axis focus point 2: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals using half harmonics
0x7F	AY_ENE RGY_POI NT3	Y-axis focus point 3: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (first harmonic)
0x80	AY_ENE RGY_POI NT4	Y-axis focus point 4: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (second harmonics)
0x81	AY_ENE RGY_POI NT5	Y-axis focus point 5: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (third harmonic)
0x82	AY_ENE RGY_POI NT6	Y-axis focus point 6: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fourth harmonic)

0x83	AY_ENE RGY_POI NT7	Y-axis focus point 7: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fifth harmonic)
0x84	AY_ENE RGY_POI NT8	Y-axis focus point 8: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (sixth harmonic)
0x85	AY_ENE RGY_AR EA1	Y-axis focus band 1 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 0 to 1/2 harmonic.
0x86	AY_ENE RGY_AR EA2	Y-axis focus band 2 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 1/2 to 1 harmonic.
0x87	AY_ENE RGY_AR EA3	Y-axis focus band 3 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band (1st to 2nd harmonic)
0x88	AY_ENE RGY_AR EA4	Y-axis focus band 4 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band at 2-3 harmonics
0x89	AY_ENE RGY_AR EA5	Y-axis focus on the acceleration spectrum energy of band 5	Energy monitoring of acceleration signals in the frequency band at 3-4 harmonics
0x8A	AY_ENE RGY_AR EA6	Y-axis focus on the acceleration spectrum energy of band 6	Energy monitoring of acceleration signals in the frequency band at 4-5 harmonics
0x8B	AY_ENE RGY	Total energy of Y acceleration spectrum	The sum of acceleration energy across all frequency ranges
0x8C	AZ_ENE RGY_POI NT1	Z-axis focus point 1: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals at quarter harmonics
0x8D	AZ_ENE RGY_POI NT2	Z-axis focus point 2: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals using half harmonics
0x8E	AZ_ENE RGY_POI NT3	Z-axis focus point 3: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (first harmonic)
0x8F	AZ_ENE	Z-axis focus point 4: Acceleration	Energy analysis of key frequency points of

	RGY_POI NT4	spectrum energy	acceleration signals (second harmonics)
0x90	AZ_ENE RGY_POI NT5	Z-axis focus point 5: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (third harmonic)
0x91	AZ_ENE RGY_POI NT6	Z-axis focus point 6: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fourth harmonic)
0x92	AZ_ENE RGY_POI NT7	Z-axis focus point 7: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fifth harmonic)
0x93	AZ_ENE RGY_POI NT8	Z-axis focus point 8: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (sixth harmonic)
0x94	AZ_ENE RGY_AR EA1	Z-axis focus band 1 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 0 to 1/2 harmonic.
0x95	AZ_ENE RGY_AR EA2	Z-axis focus band 2 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 1/2 to 1 harmonic.
0x96	AZ_ENE RGY_AR EA3	Z-axis focus on the acceleration spectrum energy of band 3	Energy monitoring of acceleration signals in the frequency band (1st to 2nd harmonic)
0x97	AZ_ENE RGY_AR EA4	Z-axis focus on the acceleration spectrum energy of band 4	Energy monitoring of acceleration signals in the frequency band at 2-3 harmonics
0x98	AZ_ENE RGY_AR EA5	Z-axis focus zone 5 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band at 3-4 harmonics

0x99	AZ_ENE RGY_AR EA6	Z-axis focus on acceleration spectrum energy (band 6)	Energy monitoring of acceleration signals in the frequency band at 4-5 harmonics
0x9A	AZ_ENE RGY	Total energy of the Z-acceleration spectrum	The sum of acceleration energy across all frequency ranges

6.4. Register Description

All the following examples are instructions for when the Modbus address is 0x50 (default). If you change the Modbus address, you need to change the address and CRC check bit in the instruction accordingly.

6.4.1. SAVE (Save/Reboot/Reset to Factory)

Register Name: SAVE

Register address: 0 (0x00)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	SAVE[15:0]	Save: 0x0000 Reboot: 0x00FF Factory Reset: 0x0001

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 00 00 FF C4 0B (reboot)

Return: 50 06 00 00 00 FF C4 0B

6.4.2. BAUD (serial port baud rate)

Register Name: BAUD

Register address: 4 (0x04)

Read/write direction: R/W

Default value: 0x0002

Bit	NAME	FUNCTION
15:4		
3:0	BAUD[3:0]	Set the serial port baud rate: 0001(0x01): 4800bps 0010(0x02): 9600bps 0011(0x03): 19200bps 0100(0x04): 38400bps 0101(0x05): 57600bps 0110(0x06): 115200bps 0111(0x07): 230400bps

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 04 00 06 45 88 (set the serial port baud rate to 115200)

Send: 50 06 00 00 00 00 84 4B (saved at 115200 baud)

Return: 50 06 00 00 00 00 84 4B

6.4.3. IICADDR (device address)

Register Name: IICADDR

Register Address: 26 (0x1A)

Read/write direction: R/W

Default value: 0x0050

Bit	NAME	FUNCTION
15:8		
7:0	IICADDR[7:0]	Set the device address, used for I2C and Modbus communication, use 0x01~0x7F

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 1A **00 02** 24 4D (set device address to **0x02**)

Return: 50 06 00 1A 00 02 24 4D

Send: **02** 06 00 69 B5 88 2F 13 (valid within 10 seconds after unlocking)

Return: 02 06 00 69 B5 88 2F 13

Send: **02** 06 00 00 00 00 89 F9 (Save)

Return: 02 06 00 00 00 00 89 F9

6.4.4. YYMM~MS (on-chip time)

Register name: YYMM~MS

Register address: 48~51 (0x30~0x33)

Read/write direction: R/W

Default time: (2015, 1, 1, 00, 00, 59, 00)

Bit	NAME	FUNCTION
15:8	YYMM[15:8]	moon
7:0	YYMM[7:0]	Year
15:8	DDHH[15:8]	hour
7:0	DDHH[7:0]	day
15:8	MMSS[15:8]	Second
7:0	MMSS[7:0]	point
15:0	MS[15:0]	millisecond

Example:

Send: 50 06 00 69 B5 88 22 A1 (valid within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 30 03 16 05 7A (set year and month 22-03)

Return: 50 06 00 30 03 16 05 7A

Send: 50 06 00 31 09 0C D3 D1 (set date and time 12-09)

Return: 50 06 00 31 09 0C D3 D1

Send: 50 06 00 32 3A 1E B7 2C (set minutes and seconds to 30:58)

Return: 50 06 00 32 3A 1E B7 2C

Send: 50 06 00 33 01 F4 74 53 (set milliseconds to 500)

Return: 50 06 00 33 01 F4 74 53

Send: 50 06 00 00 00 00 84 4B (save)

Return: 50 06 00 00 00 00 84 4B

6.4.5. AX~AZ (acceleration)

Register name: AX~AZ

Register address: 52~54 (0x34~0x36)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AX[15:0]	Acceleration X = $AX[15:0]/32768*16g$ (g is the acceleration due to gravity, which can be 9.8m/s ²)
15:0	AY[15:0]	Acceleration Y = $AY[15:0]/32768*16g$ (g is the acceleration due to gravity, which can be 9.8m/s ²)
15:0	AZ[15:0]	Acceleration Z = $AZ[15:0]/32768*16g$ (g is the acceleration due to gravity, which can be 9.8m/s ²)

Example:

Send: 50 03 00 34 00 03 49 84 (read three-axis acceleration)

Return: 50 03 06 AXH AXL AYH AYL AZH AZL CRCH CRCL

AX[15:0]=((short)AXH <<8)|AXL;

AY[15:0]=((short)AYH <<8)|AYL;

AZ[15:0]=((short)AZH <<8)|AZL;

6.4.6. VX~VZ (vibration speed)

Register name: VX~VZ

Register address: 58~60 (0x3A~0x3C)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	VX[15:0]	Vibration velocity VX (mm/s) = ((VXH << 8) VXL)/100
15:0	VY[15:0]	Vibration velocity VY (mm/S) = ((VYH << 8) VYL)/100
15:0	VZ[15:0]	Vibration velocity VZ (mm/s) = ((VZH << 8) VZL)/100

Example:

Send: 50 03 00 3A 00 03 28 47 (read the three-axis vibration speed)

Return: 50 03 06 VXH VXL VYH VYL VZH VZL CRCH CRCL

VX[15:0]=(((short)VXH <<8)|VXL)/100;

VY[15:0]=(((short)VYH <<8)|VYL)/100;

VZ[15:0]=(((short)VZH <<8)|VZL)/100;

6.4.7. Reserve

Register Name: Reserved

Register address: 61~63 (0x3D~0x3F)

Read/write direction: R

Default value: 0x0000

6.4.8. TEMP (temperature)

Register Name: TEMP

Register address: 64 (0x40)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	TEMP[15:0]	Temperature = TEMP[15:0]/100°C

Example:

Send: 50 03 00 40 00 01 88 5F (read chip temperature)

Return: 50 03 02 TEMPH TEMPL CRCH CRCL

TEMP[15:0]=(((short)TEMPH <<8)|TEMPL);

6.4.9. DX~DZ (vibration displacement)

Register name: DX~DZ

Register address: 65~67 (0x41~0x43)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	DX[15:0]	Vibration displacement DX (um) = ((DXH << 8) DXL)
15:0	DY[15:0]	Vibration displacement DY (um) = ((DYH << 8) DYL)
15:0	DZ[15:0]	Vibration displacement DZ (um) = ((DZH << 8) DZL)

Example: 60000um-1um resolution (displacement range mode)

Send: 50 03 00 41 00 03 58 5E (Read triaxial vibration displacement)

Return: 50 03 06 DXH DXL DYH DYL DZH DZL CRCH CRCL

$DX[15:0] = (((\text{short})DXH \ll 8) | DXL);$

$DY[15:0] = (((\text{short})DYH \ll 8) | DYL);$

$DZ[15:0] = (((\text{short})DZH \ll 8) | DZL);$

Example: 600um-0.01um resolution (displacement range mode)

Send: 50 03 00 41 00 03 58 5E (Read triaxial vibration displacement)

Return: 50 03 06 DXH DXL DYH DYL DZH DZL CRCH CRCL

$DX[15:0] = (((\text{short})DXH \ll 8) | DXL) / 100;$

$DY[15:0] = (((\text{short})DYH \ll 8) | DYL) / 100;$

$DZ[15:0] = (((\text{short})DZH \ll 8) | DZL) / 100;$

6.4.10. HZX ~ HZZ (vibration frequency)

Register name: HZX~HZZ

Register address: 68~70 (0x44~0x46)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	HZX[15:0]	Vibration frequency $HZX(\text{Hz}) = ((HXXH \ll 8) HXXL) / 10$
15:0	HZY[15:0]	Vibration frequency $HZY(\text{Hz}) = ((HZZH \ll 8) HZZL) / 10$
15:0	HZZ[15:0]	Vibration frequency $HZZ(\text{Hz}) = ((HZZH \ll 8) HZZL) / 10$

Example:

Send: 50 03 00 44 00 03 48 5F (read the three-axis vibration frequency)

Return: 50 03 06 HZXH HZXL HZYH HZYL HZZH HZZL CRCH CRCL

$HZX[15:0] = (((\text{short})HZXH \ll 8) | HZXL) / 10;$

$HZY[15:0] = (((\text{short})HZYH \ll 8) | HZYL) / 10;$

$HZZ[15:0] = (((\text{short})HZZH \ll 8) | HZZL) / 10;$

6.4.11. Displacement range mode

Register Name: DRM

Registry Address: 43 (0x2B)

Read/Write Direction: R/W

Default Value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	DRM[1:0]	Set displacement range mode: 0001(0x01): 60000um-1um resolution 0010(0x02): 600um-0.01um resolution

Example:

Send: 50 06 00 69 B5 88 22 A1 (Useful within 10 seconds of unlocking)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 2B 00 02 75 82 (Sets resolution from 600um to 0.01um)

Return: 50 06 00 2B 00 02 75 82

Send: 50 06 00 00 00 00 84 4B

Return: 50 06 00 00 00 00 84 4B

6.4.12. Sampling rate

Register Name: SR

Registry Address: 41 (0x29)

Read/Write Direction: R/W

Default Value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	SR [1:0]	Set sampling rate: 0000(0x00): 32K - Measurable frequency 128~16KHz 0001(0x01): 16K - Measurable frequency 64~8KHz 0010(0x02): 8K - Measurable frequency 32~4KHz 0011(0x03): 4K - Measurable frequency 16~2KHz 0100(0x04): 2K - Measurable frequency 8~1KHz 0101(0x05): 1K - Measurable frequency 4~512Hz 0110(0x06): 512Hz - Measurable frequency 2~256Hz 0111(0x07): 256Hz - Measurable frequency 1~128Hz 1000(0x08): 128Hz - Measurable frequency 1~64Hz 1001(0x09): 64Hz - Measurable frequencies 0.1~32Hz

6.4.13. Original Acceleration FIFO

Register Name: RAWFIFO

Registry Address: 44 (0x2C)

Read/Write Direction: R/W

Default Value: 0x0000

Bit	NAME	FUNCTION
15:4		

3:0	RAWFIFO[1:0]
-----	--------------

Example:

Send: 50 03 00 2C 00 01 48 42

Return: Returns raw accelerometer data, buffered with 1024 packets.

6.4.14. X-axis vibration simulation quantity

Register name: CFX~DRMSX

Register address: 71~82 (0x47~0x52)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	CFX[15:0]	X-axis acceleration peak factor $CFX(g) = ((CFXH \ll 8) \mid CFXL)/1000$
15:0	KX[15:0]	X-axis acceleration kurtosis value $KX(g) = ((KXH \ll 8) \mid KXL)/1000$
15:0	AAVGX[15:0]	The absolute average value of the X-axis acceleration, $AAVGX(g)$, is calculated as follows: $AAVGXH \ll 8 \mid AAVGXL)/1000$
15:0	VARX[15:0]	variance of X-axis acceleration, $VARX(g)$, is calculated as: $VARXH \ll 8 \mid VARXL)/1000$.
15:0	RRAX[15:0]	X-axis acceleration root mean (RMS) $RRAX(g) = ((RRAXH \ll 8 \mid RRAXL)/1000)$
15:0	WIX[15:0]	X-axis acceleration waveform index $WIX = ((WIXH \ll 8 \mid WIXL)/1000)$
15:0	PIX[15:0]	X-axis acceleration peak index $PIX(g) = ((PIXH \ll 8 \mid PIXL)/1000)$
15:0	PCX[15:0]	X-axis acceleration impulse coefficient $PCX = ((PCXH \ll 8 \mid PCXL)/1000)$
15:0	SKX[15:0]	X-axis acceleration skew $SKX(g) = ((SKXH \ll 8 \mid SKXL)/1000)$
15:0	VRMSX[15:0]	X-axis velocity RMS value $VRMSX(mm/s) = ((VRMSXH \ll 8 \mid VRMSXL)/1000)$
15:0	VKX[15:0]	X-axis velocity kurtosis value $VKX(g) = ((VKXH \ll 8 \mid VKXL) / 1000)$
15:0	DRMSX[15:0]	X-axis displacement RMS value $DRMSX(mm) = ((DRMSXH \ll 8 \mid DRMSXL) / 1000)$

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL)/1000;

Example:

Send: 50 03 00 47 00 01 39 9E (X-axis acceleration peak factor)

Return: 50 03 06 CFXH CFXL CRCH CRCL

CFX[15:0]=(((short)CFXH <<8)|CFXL)/1000;

6.4.15. Y-axis vibration simulation quantity

Register name: CFY~DRMSY

Register address: 83~94 (0x53~0x5E)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	CFY[15:0]	Y-axis acceleration peak factor CFY(g) = ((CFYH << 8) CFYL)/1000
15:0	KY[15:0]	Y-axis acceleration kurtosis value KY(g) = ((KYH << 8) KYL)/1000
15:0	AAVGY[15:0]	The absolute average value of the Y-axis acceleration, AAVGY(g), is calculated as follows: AAVGYH << 8 AAVGYL)/1000
15:0	VARY[15:0]	variance of Y-axis acceleration, VARY(g), is calculated as: VARYH << 8 VARYL)/1000.
15:0	RRAY[15:0]	Y-axis acceleration root mean (RMS) RRAY(g) = ((RRAYH << 8 RRAYL)/1000)
15:0	WIY[15:0]	Y-axis acceleration waveform index WIY = ((WIYH << 8 WIYL)/1000)
15:0	PIY[15:0]	Y-axis acceleration peak index PIY(g) = ((PIYH << 8 PIYL)/1000)
15:0	PCY[15:0]	Y-axis acceleration impulse coefficient PCY = ((PCYH << 8 PCYL)/1000)
15:0	SKY[15:0]	Y-axis acceleration skew SKY(g) = ((SKYH << 8 SKYL)/1000)

15:0	VRMSY[15:0]	Y-axis velocity RMS value VRMSY(mm/s)=((VRMSYH<< 8 VRMSYL)/1000
15:0	VKY[15:0]	Y-axis velocity kurtosis value VKY (g) = ((VKYH << 8 VKYL) / 1000
15:0	DRMSY[15:0]	Y-axis displacement RMS value DRMSY (mm) = ((DRMSYH << 8 DRMSYL) / 1000

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL)/1000;

Example:

Send: 50 03 00 47 00 01 39 9E (Y-axis acceleration peak factor)

Return: 50 03 06 CFYH CFYL CRCH CRCL

CFY[15:0]=(((short)CFYH <<8)|CFYL)/1000;

6.4.16. XZ-axis vibration simulation quantity

Register name: CFZ~DRMSZ		
Register address: 95~106 (0x5F~0x6A)		
Read/write direction: R		
Default value: 0x0000		
Bit	NAME	FUNCTION
15:0	CFZ[15:0]	Z-axis acceleration peak factor CFZ(g) = ((CFZH << 8) CFZL)/1000
15:0	KZ[15:0]	Z-axis acceleration kurtosis value KZ(g) = ((KZH << 8) KZL)/1000
15:0	AAVGZ[15:0]	The absolute average value of the Z-axis acceleration, AAVGZ(g), is calculated as follows: AAVGZH << 8 AAVGZL)/1000
15:0	VARZ[15:0]	variance of Z-axis acceleration, VARZ(g), is calculated as: VARZH << 8 VARZL)/1000.
15:0	RRAZ[15:0]	Z-axis acceleration root mean (RMS) RRAZ(g) = ((RRAZH << 8 RRAZL)/1000)

15:0	WIZ[15:0]	Z-axis acceleration waveform index $WIZ = ((WIZH < 8 \mid WIZL)/1000)$
15:0	PIZ[15:0]	Z-axis acceleration peak index $PIZ(g) = ((PIZH < 8 \mid PIZL)/1000)$
15:0	PCZ[15:0]	Z-axis acceleration impulse coefficient $PCZ = ((PCZH < 8 \mid PCZL)/1000)$
15:0	SKZ[15:0]	Z-axis acceleration skew $SKZ(g) = ((SKZH < 8 \mid SKZL)/1000)$
15:0	VRMSZ[15:0]	Z-axis velocity RMS value $VRMSZ(mm/s) = ((VRMSZH < 8 \mid VRMSZL)/1000)$
15:0	VKZ[15:0]	Z-axis velocity kurtosis value $VKZ(g) = ((VKZH < 8 \mid VKZL) / 1000)$
15:0	DRMSZ[15:0]	Z-axis displacement RMS value $DRMSZ(mm) = ((DRMSZH < 8 \mid DRMSZL) / 1000)$

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

$NAME[15:0] = (((short)NAMEH < 8) \mid NAMEL)/1000;$

Example:

Send: 50 03 00 47 00 01 39 9E (Z-axis acceleration peak factor)

Return: 50 03 06 CFZH CFZL CRCH CRCL

$CFZ[15:0] = (((short)CFZH < 8) \mid CFZL)/1000;$

6.4.17. XYZ axis fault diagnosis

Register name: ERRXERRZ

Register address: 107~109 (0x6B~0x6D)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	ERRX	
15:0	ERRY	

15:0 ERRZ

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6B 00 01 F8 57

Return: 50 03 06 ERRXH ERRXL CRCH CRCL

ERRX[15:0]=(((short)ERRXH <<8)|ERRXL);

6.4.18. X-axis focus and acc spectrum energy

Register name: AX_ENERGY_POINT1~AX_ENERGY

Register address: 110~124 (0x6E~0x7C)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AX_ENERGY_POINT1[15:0]	X-axis focus point 1: Acceleration spectrum energy AX_ENERGY_POINT1 = ((AX_ENERGY_POINT1H << 8) AX_ENERGY_POINT1L)
15:0	AX_ENERGY_POINT2[15:0]	X-axis focus point 2: Acceleration spectrum energy AX_ENERGY_POINT2 = ((AX_ENERGY_POINT2H << 8) AX_ENERGY_POINT2L)
15:0	AX_ENERGY_POINT3[15:0]	X-axis focus point 3: Acceleration spectrum energy AX_ENERGY_POINT3 = ((AX_ENERGY_POINT3H << 8) AX_ENERGY_POINT3L)
15:0	AX_ENERGY_POINT4[15:0]	X-axis focus point 4: Acceleration spectrum energy AX_ENERGY_POINT4 = ((AX_ENERGY_POINT4H << 8) AX_ENERGY_POINT4L)
15:0	AX_ENERGY_POINT5[15:0]	X-axis focus point 5: Acceleration spectrum energy AX_ENERGY_POINT5 = ((AX_ENERGY_POINT5H << 8) AX_ENERGY_POINT5L)
15:0	AX_ENERGY_POINT6[15:0]	X-axis focus point 6: Acceleration spectrum energy AX_ENERGY_POINT6 = ((AX_ENERGY_POINT6H << 8) AX_ENERGY_POINT6L)

15:0	AX_ENERGY_POINT7[15:0]	X-axis focus point 7: Acceleration spectrum energy AX_ENERGY_POINT7 = ((AX_ENERGY_POINT7H << 8) AX_ENERGY_POINT7L)
15:0	AX_ENERGY_POINT8[15:0]	X-axis focus point 8: Acceleration spectrum energy AX_ENERGY_POINT8 = ((AX_ENERGY_POINT8H << 8) AX_ENERGY_POINT8L)
15:0	AX_ENERGY_AREA1[15:0]	X-axis focus band 1 acceleration spectrum energy AX_ENERGY_AREA1=((AX_ENERGY_AREA1H<< 8 AX_ENERGY_AREA1L)
15:0	AX_ENERGY_AREA2[15:0]	X-axis focus band 2 acceleration spectrum energy AX_ENERGY_AREA2=((AX_ENERGY_AREA2H<< 8 AX_ENERGY_AREA2L)
15:0	AX_ENERGY_AREA3[15:0]	X-axis focus band 3 acceleration spectrum energy AX_ENERGY_AREA3=((AX_ENERGY_AREA3H<< 8 AX_ENERGY_AREA3L)
15:0	AX_ENERGY_AREA4[15:0]	X-axis focus band 4 acceleration spectrum energy AX_ENERGY_AREA4=((AX_ENERGY_AREA4H<< 8 AX_ENERGY_AREA4L)
15:0	AX_ENERGY_AREA5[15:0]	X-axis focus band 5 acceleration spectrum energy AX_ENERGY_AREA5=((AX_ENERGY_AREA5H<< 8 AX_ENERGY_AREA5L)
15:0	AX_ENERGY_AREA6[15:0]	X-axis focus band 6 acceleration spectrum energy AX_ENERGY_AREA6=((AX_ENERGY_AREA6H<< 8 AX_ENERGY_AREA6L)
15:0	AX_ENERGY[15:0]	Total energy of the X-axis acceleration spectrum AX_ENERGY = ((AX_ENERGYH << 8 AX_ENERGYL)

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6E 00 01 E8 56 (X-axis focus point 1: Acceleration spectrum energy)

Return: 50 03 06 AX_ENERGY_POINT1H AX_ENERGY_POINT1L CRCH CRCL

AX_ENERGY_POINT1[15:0]=(((short)AX_ENERGY_POINT1H <<8)|AX_ENERGY_POINT1L);

6.4.19. Y-axis focus and acc spectrum energy

Register name: AY_ENERGY_POINT1~AY_ENERGY

Register address: 125~139 (0x7D~0x8B)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AY_ENERGY_POINT1[15:0]	Y-axis focus point 1: Acceleration spectrum energy AY_ENERGY_POINT1 = ((AY_ENERGY_POINT1H << 8) AY_ENERGY_POINT1L)
15:0	AY_ENERGY_POINT2[15:0]	Y-axis focus point 2: Acceleration spectrum energy AY_ENERGY_POINT2 = ((AY_ENERGY_POINT2H << 8) AY_ENERGY_POINT2L)
15:0	AY_ENERGY_POINT3[15:0]	Y-axis focus point 3: Acceleration spectrum energy AY_ENERGY_POINT3 = ((AY_ENERGY_POINT3H << 8) AY_ENERGY_POINT3L)
15:0	AY_ENERGY_POINT4[15:0]	Y-axis focus point 4: Acceleration spectrum energy AY_ENERGY_POINT4 = ((AY_ENERGY_POINT4H << 8) AY_ENERGY_POINT4L)
15:0	AY_ENERGY_POINT5[15:0]	Y-axis focus point 5: Acceleration spectrum energy AY_ENERGY_POINT5 = ((AY_ENERGY_POINT5H << 8) AY_ENERGY_POINT5L)
15:0	AY_ENERGY_POINT6[15:0]	Y-axis focus point 6: Acceleration spectrum energy AY_ENERGY_POINT6 = ((AY_ENERGY_POINT6H << 8) AY_ENERGY_POINT6L)
15:0	AY_ENERGY_POINT7[15:0]	Y-axis focus point 7: Acceleration spectrum energy AY_ENERGY_POINT7 = ((AY_ENERGY_POINT7H << 8) AY_ENERGY_POINT7L)
15:0	AY_ENERGY_POINT8[15:0]	Y-axis focus point 8: Acceleration spectrum energy AY_ENERGY_POINT8 = ((AY_ENERGY_POINT8H << 8) AY_ENERGY_POINT8L)
15:0	AY_ENERGY_AREA1[15:0]	Y-axis focus band 1 acceleration spectrum energy AY_ENERGY_AREA1=((AY_ENERGY_AREA1H<< 8 AY_ENERGY_AREA1L)
15:0	AY_ENERGY_AREA2[15:0]	Y-axis focus band 2 acceleration spectrum energy AY_ENERGY_AREA2=((AY_ENERGY_AREA2H<< 8 AY_ENERGY_AREA2L)
15:0	AY_ENERGY_AREA3[15:0]	Y-axis focus band 3 acceleration spectrum energy AY_ENERGY_AREA3=((AY_ENERGY_AREA3H<< 8 AY_ENERGY_AREA3L)
15:0	AY_ENERGY_AREA4[15:0]	Y-axis focus band 4 acceleration spectrum energy AY_ENERGY_AREA4=((AY_ENERGY_AREA4H<< 8 AY_ENERGY_AREA4L)
15:0	AY_ENERGY_AREA5[15:0]	Y-axis focus band 5 acceleration spectrum energy AY_ENERGY_AREA5=((AY_ENERGY_AREA5H<< 8 AY_ENERGY_AREA5L)
15:0	AY_ENERGY_AREA6[15:0]	Y-axis focus band 6 acceleration spectrum energy AY_ENERGY_AREA6=((AY_ENERGY_AREA6H<< 8 AY_ENERGY_AREA6L)
15:0	AY_ENERGY[15:0]	Total energy of the Y-axis acceleration spectrum AY_ENERGY = ((AY_ENERGYH << 8 AY_ENERGYL)

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6E 00 01 E8 56 (Y-axis focus point 1: Acceleration spectrum energy)

Return: 50 03 06 AY_ENERGY_POINT1H AY_ENERGY_POINT1L CRCH CRCL

AY_ENERGY_POINT1[15:0]=(((short)AY_ENERGY_POINT1H <<8)|AY_ENERGY_POINT1L);

6.4.20. Z-axis focus and acc spectrum energy

Register name: AZ_ENERGY_POINT1~AZ_ENERGY

Register address: 140~154 (0x8C~0x9A)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AZ_ENERGY_POINT1[15:0]	Z-axis focus point 1: Acceleration spectrum energy AZ_ENERGY_POINT1 = ((AZ_ENERGY_POINT1H << 8) AZ_ENERGY_POINT1L)
15:0	AZ_ENERGY_POINT2[15:0]	Z-axis focus point 2: Acceleration spectrum energy AZ_ENERGY_POINT2 = ((AZ_ENERGY_POINT2H << 8) AZ_ENERGY_POINT2L)
15:0	AZ_ENERGY_POINT3[15:0]	Z-axis focus point 3: Acceleration spectrum energy AZ_ENERGY_POINT3 = ((AZ_ENERGY_POINT3H << 8) AZ_ENERGY_POINT3L)
15:0	AZ_ENERGY_POINT4[15:0]	Z-axis focus point 4: Acceleration spectrum energy AZ_ENERGY_POINT4 = ((AZ_ENERGY_POINT4H << 8) AZ_ENERGY_POINT4L)
15:0	AZ_ENERGY_POINT5[15:0]	Z-axis focus point 5: Acceleration spectrum energy AZ_ENERGY_POINT5 = ((AZ_ENERGY_POINT5H << 8) AZ_ENERGY_POINT5L)
15:0	AZ_ENERGY_POINT6[15:0]	Z-axis focus point 6: Acceleration spectrum energy AZ_ENERGY_POINT6 = ((AZ_ENERGY_POINT6H << 8) AZ_ENERGY_POINT6L)
15:0	AZ_ENERGY_POINT7[15:0]	Z-axis focus point 7: Acceleration spectrum energy AZ_ENERGY_POINT7 = ((AZ_ENERGY_POINT7H << 8) AZ_ENERGY_POINT7L)

15:0	AZ_ENERGY_POINT8[15:0]	Z-axis focus point 8: Acceleration spectrum energy AZ_ENERGY_POINT8 = ((AZ_ENERGY_POINT8H << 8) AZ_ENERGY_POINT8L)
15:0	AZ_ENERGY_AREA1[15:0]	Z-axis focus band 1 acceleration spectrum energy AZ_ENERGY_AREA1=((AZ_ENERGY_AREA1H<< 8 AZ_ENERGY_AREA1L)
15:0	AZ_ENERGY_AREA2[15:0]	Z-axis focus band 2 acceleration spectrum energy AZ_ENERGY_AREA2=((AZ_ENERGY_AREA2H<< 8 AZ_ENERGY_AREA2L)
15:0	AZ_ENERGY_AREA3[15:0]	Z-axis focus band 3 acceleration spectrum energy AZ_ENERGY_AREA3=((AZ_ENERGY_AREA3H<< 8 AZ_ENERGY_AREA3L)
15:0	AZ_ENERGY_AREA4[15:0]	Z-axis focus band 4 acceleration spectrum energy AZ_ENERGY_AREA4=((AZ_ENERGY_AREA4H<< 8 AZ_ENERGY_AREA4L)
15:0	AZ_ENERGY_AREA5[15:0]	Z-axis focus band 5 acceleration spectrum energy AZ_ENERGY_AREA5=((AZ_ENERGY_AREA5H<< 8 AZ_ENERGY_AREA5L)
15:0	AZ_ENERGY_AREA6[15:0]	Z-axis focus band 6 acceleration spectrum energy AZ_ENERGY_AREA6=((AZ_ENERGY_AREA6H<< 8 AZ_ENERGY_AREA6L)
15:0	AZ_ENERGY[15:0]	Total energy of the Z-axis acceleration spectrum AZ_ENERGY = ((AZ_ENERGYH << 8 AZ_ENERGYL)

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6E 00 01 E8 56 (Z-axis focus point 1: Acceleration spectrum energy)

Return: 50 03 06 AZ_ENERGY_POINT1H AZ_ENERGY_POINT1L CRCH CRCL

AZ_ENERGY_POINT1[15:0]=(((short)AZ_ENERGY_POINT1H <<8)|AZ_ENERGY_POINT1L);